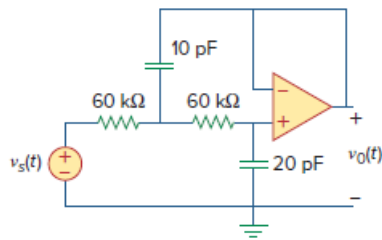


Problem

Obtain V_0/V_s in the op amp circuit in Fig. 16.91.

Figure 16.91:



Step-by-step solution

Step 1 of 4

Refer to Figure 16.91 from the textbook.

Transform the circuit in Figure 16.91 to frequency domain.

The Laplace transform of $v_s(t)$ is $V_s(s)$ and $v_o(t)$ is $V_o(s)$.

The Laplace transform of capacitance 10 pF is $\frac{10^{11}}{s}$ and 20 pF is $\frac{5(10^{10})}{s}$.

The frequency domain circuit is shown in Figure 1.

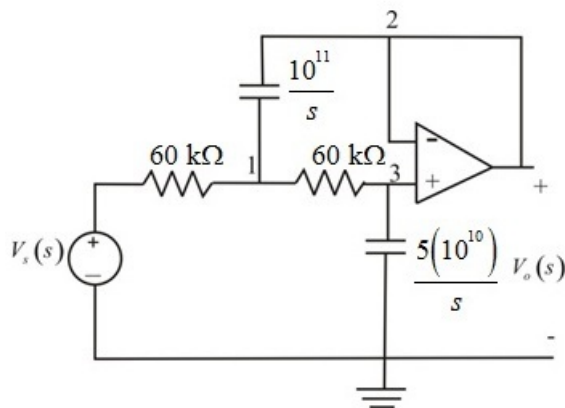


Figure 1

Step 2 of 4

Assume voltage at node 1 is $V_1(s)$, voltage at node 3 is $V_3(s)$, voltage at the node 2 is $V_2(s)$ and is equal to output voltage $V_o(s)$.

$$V_2(s) = V_o(s)$$

Apply Kirchhoff's current law at node 1.

$$\frac{V_1(s) - V_s(s)}{60(10^3)} + \left(\frac{V_1(s) - V_o(s)}{\frac{10^{11}}{s}} \right) + \frac{V_1(s) - V_3(s)}{60(10^3)} = 0$$

$$V_1(s) \left[\frac{1}{60(10^3)} + s10^{-11} + \frac{1}{60(10^3)} \right] - 10^{-11}sV_o(s) = \frac{1}{60(10^3)} [V_s(s) + V_3(s)]$$

$$V_1(s) \left[\frac{1}{30(10^3)} + s10^{-11} \right] - 10^{-11}sV_o(s) = \frac{1}{60(10^3)} [V_s(s) + V_3(s)] \dots\dots (1)$$

Step 3 of 4

Apply Kirchhoff's current law at node 2.

$$\frac{V_2(s) - V_1(s)}{\frac{10^{11}}{s}} = 0$$

$$V_o(s) - V_1(s) = 0$$

$$V_1(s) = V_o(s) \dots\dots (2)$$

Apply Kirchhoff's current law at node 3.

$$\frac{V_3(s) - V_1(s)}{60(10^3)} + \left(\frac{V_3(s) - 0}{\frac{5(10^{10})}{s}} \right) = 0$$

$$V_3 \left[\frac{1}{60(10^3)} + \frac{s}{5(10^{10})} \right] - \frac{V_1(s)}{60(10^3)} = 0$$

$$V_3 \left[\frac{1}{60(10^3)} + (20)(10^{-12})s \right] = \frac{V_1(s)}{60(10^3)}$$

Step 4 of 4

From equation (2).

$$V_3(s) \left[\frac{1}{60(10^3)} + (20)(10^{-12})s \right] = \frac{V_o(s)}{60(10^3)}$$

$$V_3(s) \left[\frac{1 + (12 \times 10^{-7})s}{60(10^3)} \right] = \frac{V_o(s)}{60(10^3)}$$

$$V_3(s) = V_o(s) \left[\frac{1}{1 + (12 \times 10^{-7})s} \right] \dots\dots (3)$$

Substitute equations (2) and (3) in equation (1).

$$V_o(s) \left[\frac{1}{30(10^3)} + 10^{-11}s \right] - 10^{-11}s V_o(s) = \frac{1}{60(10^3)} \left[V_s(s) + \frac{V_o(s)}{1 + (12 \times 10^{-7})s} \right]$$

$$V_o(s) \left[\frac{1}{30(10^3)} \right] = \frac{V_s(s)}{60(10^3)} + V_o(s) \left(\frac{1}{60(10^3) + 60(10^3)(12 \times 10^{-7})s} \right)$$

Continue the calculations.

$$V_o(s) \left[\frac{60 + (720 \times 10^{-7})s - 30}{1800 + (216 \times 10^{-5})s} \right] = \frac{V_s(s)}{60}$$

$$\frac{V_o(s)}{V_s(s)} = \left(\frac{1}{60} \right) \left[\frac{1800 + (216 \times 10^{-5})s}{30 + (720 \times 10^{-7})s} \right]$$

$$\frac{V_o(s)}{V_s(s)} = \frac{30 + (3.6 \times 10^{-5})s}{30 + (720 \times 10^{-7})s}$$

$$\begin{aligned} \frac{V_o(s)}{V_s(s)} &= \frac{1 + (0.12 \times 10^{-5})s}{1 + (24 \times 10^{-7})s} \\ &= \frac{s + (0.12 \times 10^{-5})}{s + (24 \times 10^{-7})} \end{aligned}$$

Therefore, $\frac{V_o(s)}{V_s(s)}$ in the op amp circuit is $\boxed{\frac{s + (0.12 \times 10^{-5})}{s + (24 \times 10^{-7})}}$.